

CROSS-CUTTING · EXAM STRATEGY - ALL DOMAINS

Migration and Hybrid Architecture

Getting there, and staying partly here.

COVERS ALL DOMAINS SCENARIOS 8 MODULE 14 of 22

Hybrid is not a failure to finish migrating.
For many workloads it is the **intended
destination.**

The seven migration strategies

STRATEGY	WHAT IT MEANS	SIGNAL
Rehost	Lift and shift, unchanged	'As quickly as possible', 'minimal changes'
Replatform	Small optimisations during the move	'Move to managed database while migrating'
Repurchase	Drop it and buy a SaaS product	'Replace with a commercial product'
Refactor	Rewrite for cloud-native architecture	'Re-architect', 'take advantage of serverless'
Retire	Switch it off. Nobody uses it	'Discovered unused during assessment'
Retain	Leave it where it is, for now	'Not ready to move', 'recently invested in'
Relocate	Move hypervisor-level, e.g. VMware Cloud	'Move VMware estate without conversion'

ON THE EXAM

Rehost is fastest and captures least value. Refactor captures most and takes longest. The question tells you which matters.

The services that support a migration

- ◆ **AWS Application Discovery Service.** Inventories on-premises servers and their dependencies before you plan.
- ◆ **AWS Migration Hub.** A single view of migration progress across tools and Regions.
- ◆ **AWS Application Migration Service.** The standard rehost tool. Block-level replication, short cutover.
- ◆ **AWS DMS.** Database migration with ongoing replication. Pair with the Schema Conversion Tool across engines.
- ◆ **AWS DataSync.** File and object transfer over the network, one-off or scheduled.

ON THE EXAM

'We do not know what we have or what depends on what' is Application Discovery Service, before any migration begins.

Storage Gateway, in three modes

MODE	PRESENTS	CHOOSE IT WHEN
File Gateway	NFS or SMB file share, backed by S3	Applications need a file share; data belongs in S3
Volume Gateway	iSCSI block volumes, backed by S3 snapshots	Applications need block storage with cloud backup
Tape Gateway	A virtual tape library	Existing backup software writes to tape and will not change

ON THE EXAM

Tape Gateway has one signal: existing backup software that expects a tape library. It is the 'do not rewrite it' answer.

Bringing AWS to your premises

SERVICE	WHAT IT IS	SIGNAL
AWS Outposts	AWS racks in your data centre	'Data must stay on site'
AWS Local Zones	AWS infrastructure near a city	'Low latency for one city'
AWS Wavelength	Compute inside a 5G network	'Mobile users', 'over 5G'
Snowball Edge	Rugged compute and storage	'Disconnected', 'harsh site'
Snowcone	Small portable device	'Drone', 'vehicle'

ON THE EXAM

Outposts is your data centre. Local Zones is a city. Wavelength is a mobile network.

A company plans to migrate around four hundred servers to AWS. Nobody has an accurate inventory, several applications have undocumented dependencies on one another, and two previous attempts failed when moving one server broke something unrelated. Leadership wants the migration planned properly this time.

What should the solutions architect recommend FIRST?

- ☐ A Begin rehosting with AWS Application Migration Service, starting with the least critical servers
- ☐ B Run AWS Application Discovery Service to build an inventory and dependency map
- ☐ C Order Snowball devices to begin moving the data
- ☐ D Refactor the applications to a serverless architecture

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Start rehosting
Migrating without a dependency map is exactly what caused the two previous failures.
- B

 Application Discovery Service
Builds the inventory and dependency map the company lacks, which is the missing input for every subsequent decision.
- C

 Order Snowball
A transfer mechanism chosen before anyone knows what is being transferred or how much.
- D

 Refactor everything
The longest and most expensive path, chosen with no understanding of what the applications currently do.

THE SIGNAL

'Nobody knows what we have' means discovery comes first. The word FIRST is doing the work again.

A manufacturing company runs control software that must respond to factory equipment within five milliseconds. Regulation also requires that production data never leave the site. The company wants to use AWS services and a consistent set of APIs across its estate.

Which solution meets these requirements?

- ☐ A Deploy the workload to the nearest AWS Region with Direct Connect
- ☐ B Deploy AWS Outposts in the company's own data centre
- ☐ C Use an AWS Local Zone in the nearest metropolitan area
- ☐ D Use AWS Wavelength in the local 5G network

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A** Nearest Region plus Direct Connect
Data leaves the site, breaking the regulation, and Region round-trip cannot reliably hit five milliseconds.
- C** Local Zone
Closer than a Region but still off site, so the data residency requirement fails.

- B** Outposts on site
AWS-managed racks physically in the data centre, so data stays on site and latency to local equipment is minimal, with the same APIs.
- D** Wavelength
For mobile devices on a 5G network. Factory equipment on a local wired network is not the use case.

THE SIGNAL

'Data must never leave the site' rules out everything except Outposts. Local Zones and Wavelength are still AWS-side.

A company's backup software writes to a physical tape library that is reaching end of life. Replacing the backup software would require retraining and revalidation that the business will not fund this year.

They want the backups stored durably in AWS instead of on tape.

Which solution should be used?

- ☐ A AWS Storage Gateway in Tape Gateway mode
- ☐ B AWS Storage Gateway in File Gateway mode
- ☐ C AWS DataSync copying the tape contents nightly
- ☐ D AWS Backup configured against the on-premises servers

◆ PAUSE AND DECIDE

Which solution should be used?

- A** Tape Gateway
Presents a virtual tape library, so the existing backup software writes to what it thinks is tape while data lands in S3 and Glacier.
- B** File Gateway
Presents a file share. Backup software expecting a tape library will not write to it without reconfiguration.
- C** DataSync from tape
DataSync moves files between file and object storage. It does not read a physical tape library.
- D** AWS Backup on premises
AWS Backup manages AWS resources and some hybrid sources, and it does not preserve the existing tape-based workflow.

THE SIGNAL

Existing backup software plus tape plus unwillingness to change is the one Tape Gateway signal.

A company must move a two hundred terabyte media archive to S3 from a remote location with no reliable internet connection. The site is a temporary film production base in a desert with intermittent power.

The archive must reach AWS within a month.

Which solution is MOST appropriate?

- ☐ A AWS DataSync over a satellite internet link
- ☐ B AWS Snowball Edge devices shipped to the site
- ☐ C AWS Direct Connect to the production base
- ☐ D S3 Transfer Acceleration with multipart upload

◆ PAUSE AND DECIDE

Which solution is MOST appropriate?

- A

 DataSync over satellite
Two hundred terabytes over an intermittent satellite link will not complete in a month, if at all.
- B

 Snowball Edge
Ruggedised for harsh environments, needs no network at all, and shipping plus loading fits inside a month.
- C

 Direct Connect
A permanent circuit taking weeks to months to provision, to a temporary site in a desert.
- D

 Transfer Acceleration
Optimises use of available bandwidth. There is no reliable bandwidth here to optimise.

THE SIGNAL

Snowball is not only about volume. Disconnected, remote or harsh environments point to it regardless of size.

A company must migrate three hundred virtual machines to AWS within four months. The applications are stable, the business has no appetite for code changes during the move, and speed is the stated priority.

A consultant has proposed refactoring each application to containers as part of the migration.

What should the solutions architect advise?

- ☐ A Proceed with refactoring, since containers are the modern architecture
- ☐ B Rehost with AWS Application Migration Service, and modernise afterwards
- ☐ C Repurchase equivalent SaaS products for all three hundred applications
- ☐ D Retain the applications on premises indefinitely

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Refactor during migration

The slowest strategy, requiring exactly the code changes the business has ruled out, against a four-month deadline.

C Repurchase everything

Replacing three hundred applications with commercial products in four months is not achievable, and would change every business process.

B Rehost now, modernise later

Fastest path, no application changes, and it does not prevent refactoring afterwards once workloads are running on AWS.

D Retain on premises

Ignores the requirement to migrate.

THE SIGNAL

Speed plus no code changes means rehost. Refactoring is not wrong in general, only wrong under these constraints.

A mobile game requires end-to-end latency under ten milliseconds for players connected over 5G in three large cities. Players are on mobile devices, not fixed connections.

Regional deployment produces around forty milliseconds, which players report as unplayable.

Which service should be used?

- ☐ A AWS Wavelength Zones in the carriers' networks
- ☐ B AWS Outposts in the company's data centre
- ☐ C Amazon CloudFront with regional edge caches
- ☐ D AWS Global Accelerator to the nearest Region

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Wavelength Zones
Places compute inside the carrier's 5G network, so traffic never traverses the public internet to reach a Region.
- B

 Outposts
Puts AWS in your data centre. Mobile players would still cross the internet to reach it.
- C

 CloudFront
Caches content at the edge. Interactive game state is not cacheable.
- D

 Global Accelerator
Improves the path to a Region and still cannot get below the physical round-trip the Region imposes.

THE SIGNAL

5G plus mobile devices plus single-digit millisecond latency is the one Wavelength signal.

During a migration assessment, the team finds that of four hundred servers, sixty run applications that no department claims, have had no user logins in eighteen months, and generate no network traffic beyond monitoring.

The migration budget is under pressure.

Which strategy applies to those sixty servers?

- ☐ A Rehost them to the smallest available instance types
- ☐ B Retire them
- ☐ C Retain them on premises until an owner is found
- ☐ D Replatform them onto managed services

◆ PAUSE AND DECIDE

Which strategy applies to those sixty servers?

- A

Rehost small
Spends migration effort and ongoing money running software nobody uses.
- B

Retire
No owner, no logins for eighteen months, no traffic. Retiring removes fifteen percent of the migration scope and its ongoing cost.
- C

Retain on premises
Keeps paying for hardware and licences for unused software, and defers the decision indefinitely.
- D

Replatform
Invests engineering effort in modernising applications nobody uses.

THE SIGNAL

Discovering things to switch off is one of the main reasons to run an assessment. Retire is a real answer.

A hospital's imaging system must keep reading and writing to a local file share for performance, while the master copy of the data lives in S3 for analytics and long-term retention.

Local storage cannot hold the full archive, but clinicians must open recent studies without delay.

Which solution meets these requirements?

- ☐ A Storage Gateway in File Gateway mode
- ☐ B Storage Gateway in Volume Gateway mode
- ☐ C AWS DataSync running every hour
- ☐ D Amazon FSx for Windows File Server in the Region

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A File Gateway

Presents a local file share backed by S3, caching recent data on site so the full archive need not fit locally.

C Hourly DataSync

Copies data on a schedule without presenting a local share, and the site cannot hold the full archive anyway.

B Volume Gateway

Presents block storage over iSCSI. The application expects a file share, and data lands as snapshots rather than usable S3 objects.

D FSx in the Region

Every read crosses the network to AWS, failing the local performance requirement.

THE SIGNAL

File share plus S3 as the master copy plus local caching is File Gateway. Volume Gateway is for block, and its data is not directly usable in S3.

Module 14 takeaways

- ◆ **'Nobody knows what we have'** means Application Discovery Service, before anything moves.
- ◆ **Speed plus no code changes** means rehost. Modernise afterwards.
- ◆ **Data must never leave the site** means Outposts. Local Zones and Wavelength are still off site.
- ◆ **Existing backup software writing to tape** means Tape Gateway.
- ◆ **Retire is a real answer.** Assessments routinely find servers nobody uses.

ON THE EXAM

Next: federation, application integration, and the managed AI services that appear as service-matching questions.